

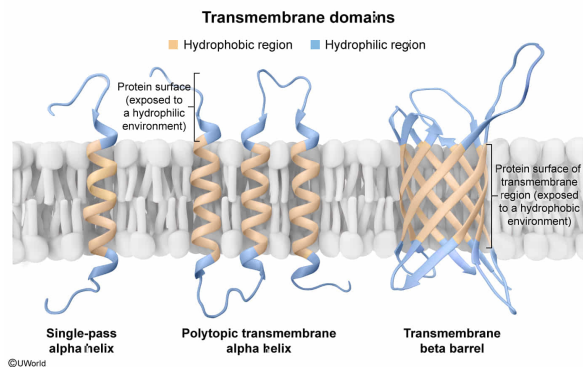
The cystic fibrosis transmembrane conductance regulator (CFTR) protein is a chloride ion channel involved in the production of mucus, sweat, and digestive fluids. CFTR is composed of five domains (Figure 1): two transmembrane domains (TMD1 and TMD2), two nucleotide binding domains (NBD1 and NBD2), and one regulatory domain (R domain).

Figure 1 Schematic diagram of CFTR domains

Cystic fibrosis (CF) arises from recessively inherited mutations in the *CFTR* gene. The most common mutation is the deletion of three base-pairs that removes a phenylalanine residue at position 508 in NBD1 ($\Delta F508$). This mutation causes the protein to fold incorrectly, making it more susceptible to degradation by proteases such as trypsin. The resulting decrease in CFTR abundance inhibits chloride ion transport, leading to thickening of normally thin secretions and increased risk of life-threatening pulmonary infections.

- ✔ ☒ A. TMD1 [77%]
☐ B. NBD1 [8%]
☐ C. R domain [10%]
☐ D. NBD2 [4%]

Explanation:



The surface of a protein is the portion of that protein that is exposed to its immediate environment, as opposed to the interior of the protein, which is not in direct contact with the surrounding environment. In most cases, a protein's surface is in contact with an aqueous environment (ie, water). However, **transmembrane proteins** cross the **phospholipid bilayer** of cellular membranes. Therefore, the surface of a transmembrane region is in contact with the fatty acid tails of the bilayer rather than water.

Fatty acid tails are hydrophobic and, as such, interact poorly with hydrophilic chemical groups. Instead, they interact favorably with hydrophobic molecules. For this reason, the surfaces of the transmembrane regions of a protein consist mostly of hydrophobic amino acid residues.

CFTR has two domains that contain transmembrane regions, TMD1 and TMD2. Therefore, **Isoleucine** (a **hydrophobic amino acid**) is likely to be found on the surface of CFTR in both TMD1 and TMD2, where the surrounding environment is *a/s*o hydrophobic. However, TMD2 is not an answer choice, so of the available options, the **TMD1 domain is most likely to have isoleucine residues on its surface** rather than buried within the protein.

(Choices B, C, and D) Figure 1 shows that both the NBD domains and the R domain of CFTR are found in the aqueous environment of the cytosol. Because hydrophobic residues interact poorly with aqueous environments, isoleucine residues in these domains are more likely to be found buried **inside the protein**, where they can interact with other hydrophobic amino acids rather than on the surface, where they would be forced to interact with water.

Educational objective:

The transmembrane domains of proteins cross the hydrophobic environment of the phospholipid bilayer. Hydrophobic amino acids are likely to be found on the surface of transmembrane regions, where they can interact with the hydrophobic environment.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402098
System:
Amino Acids and Proteins

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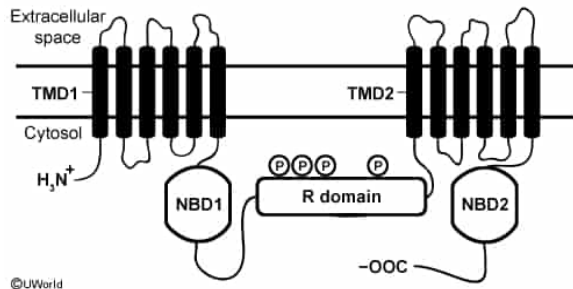


Figure 1 Schematic diagram of CFTR domains

The NBD domains regulate CFTR activity by binding ATP, which causes a conformational change in the TMD domains. This change allows chloride ions to passively cross the membrane down their concentration gradient. The movement of chloride ions facilitates the simultaneous movement of water in the same direction by osmotic pressure, which helps keep secretions thin. The R domain further regulates activity through phosphorylation at several positions.

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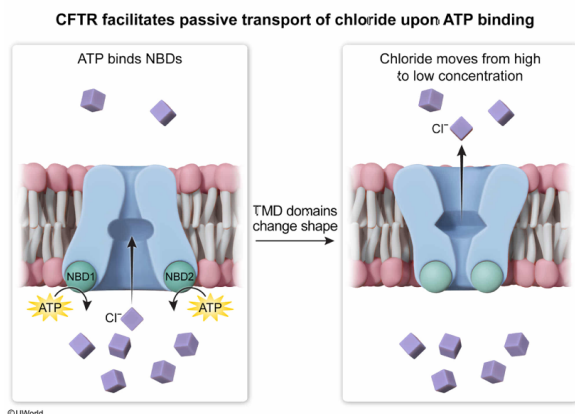
According to the information in the passage, which statement best describes the role of ATP in CFTR activity? ATP:

- ☐ A. is hydrolyzed to provide the energy input necessary for ion transport. [18%]
- ☐ B. flows through CFTR in the opposite direction as chloride. [1%]
- ☐ C. inhibits chloride transport by occupying the channel. [1%]
- ✓ ☒ D. allosterically activates CFTR-mediated chloride transport. [78%]

Correct

79% answered correctly

Explanation:



Protein **activity can be modulated** by interactions with **allosteric effectors**. Allosteric effectors **bind to proteins at one site** and cause a **change in the shape (conformational change)** of the protein at **another site**, thereby altering its ability to function. Activators increase activity, and inhibitors decrease activity.

According to the passage, the binding of ATP to the NBD domains causes a conformational change in a different part of the protein: the TMD domains. This conformational change allows chloride ions to cross the membrane. Because ATP increases the ability of CFTR to carry out its function (chloride transport), ATP acts as an activator. In addition, ATP binds at a different site on the protein than the site that is affected. Therefore, **ATP allosterically activates** chloride transport by CFTR.

(Choice A) Although ATP binding is necessary for ion transport by CFTR, the passage states that the **ions travel passively** down their concentration gradient. Energy input from the hydrolysis of ATP is only required for **active transport**, in which ions travel *against* their concentration gradient.

(Choice B) According to the passage, ATP *binds* to the NBD domains of CFTR. ATP does *not* flow through CFTR in either direction.

(Choice C) According to the passage, ATP binding *facilitates* chloride transport by opening the CFTR channel. ATP does *not* occupy the channel or inhibit transport.

Educational objective:

Allosteric effectors bind proteins at one site and induce a conformational change (change in shape) at another site. Allosteric activators increase the ability of the protein to function, and inhibitors decrease the function.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402099
System:
Amino Acids and Proteins

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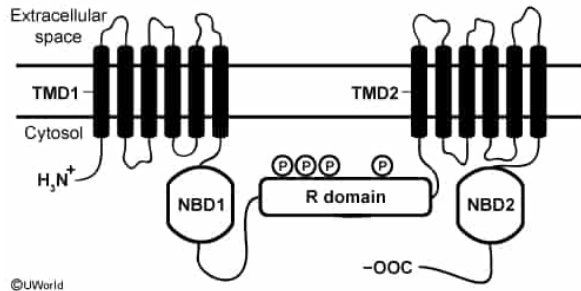


Figure 1 Schematic diagram of CFTR domains

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Cystic fibrosis (CF) arises from recessively inherited mutations in the *CFTR* gene. The most common mutation is the deletion of three base-pairs that removes a phenylalanine residue at position 508 in NBD1 ($\Delta F508$). This mutation causes the protein to fold incorrectly, making it more susceptible to degradation by proteases such as trypsin. The resulting decrease in CFTR abundance inhibits chloride ion transport, leading to thickening of normally thin secretions and increased risk of life-threatening pulmonary infections.

A scientist proposed that phosphorylation of the R domain increases CFTR activity. Which of the following functional groups in the R domain would be the best to remove by mutation to test this hypothesis *in vivo*?

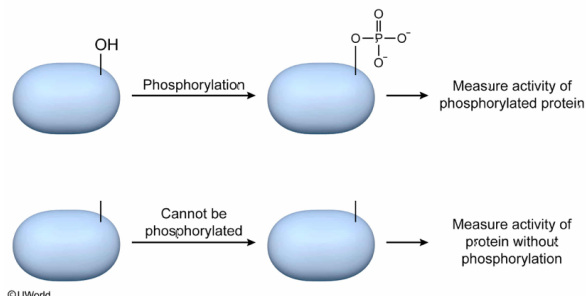
- ☐ A. Amide [14%]
- ☒ B. Hydroxyl [56%]
- ☐ C. Thiol [14%]
- ☐ D. Carboxylic acid [15%]

Correct

56% answered correctly

Explanation:

Determining the effect of phosphorylation on protein activity *in vivo*



Post-translational modifications are covalent additions of **non-amino acid groups** to a protein. They are catalyzed by enzymes and generally only occur at **specific amino acid residues**. **Phosphorylation** is a common post-translational modification that is facilitated by **protein kinase enzymes**.

Phosphorylation can occur at various residues, but by far the most common are **serine**, **threonine**, and **tyrosine** residues, each of which contains a **hydroxyl group** in its side chain. Consequently, protein kinases almost always catalyze reactions in which the hydroxyl group on an amino acid side chain nucleophilically attacks a phosphate group from ATP.

The effects of a post-translational modification (eg, phosphorylation) on protein activity can be investigated by generating a mutated form of the protein that cannot be phosphorylated. The new amino acids should **lack hydroxyl groups** to prevent phosphorylation but should otherwise be as similar to the original amino acids as possible to preserve protein structure (ie, **tyrosine** to **phenylalanine**, **serine** and **threonine** to **alanine**). Scientists can then compare the functional activity of the unphosphorylated protein to the activity of the phosphorylated protein and determine which is greater.

(Choice A) The amide groups of **asparagine** and **glutamine** may be glycosylated (usually asparagine) but are rarely phosphorylated.

(Choice C) Thiols are the functional groups of **cysteine** residues. Although cysteine can act as a nucleophile, it rarely participates in phosphorylation.

(Choice D) Carboxylic acids, found in **aspartate** and **glutamate**, are rarely phosphorylated.

Educational objective:

Post-translational modifications are covalent additions of non-amino acid groups to proteins. They generally occur at specific residues that have the necessary chemical properties. Phosphorylation occurs almost exclusively at the hydroxyl groups of serine, threonine, and tyrosine residues.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402100
System:
Amino Acids and Proteins

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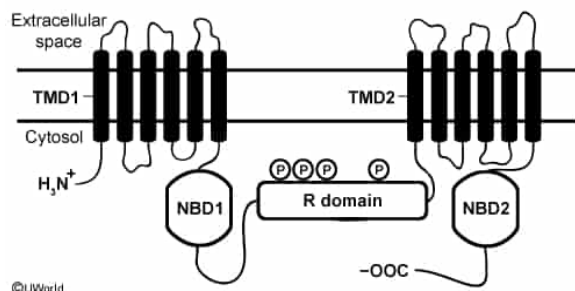


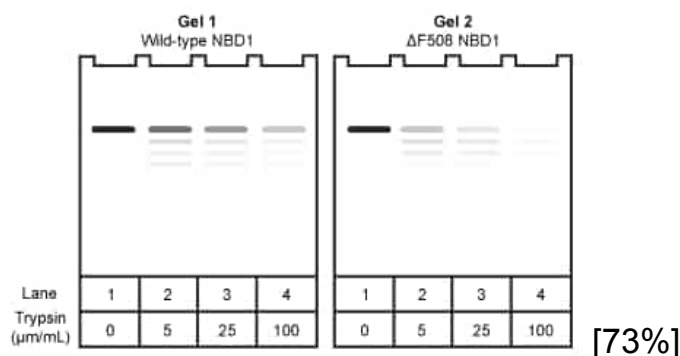
Figure 1 Schematic diagram of CFTR domains

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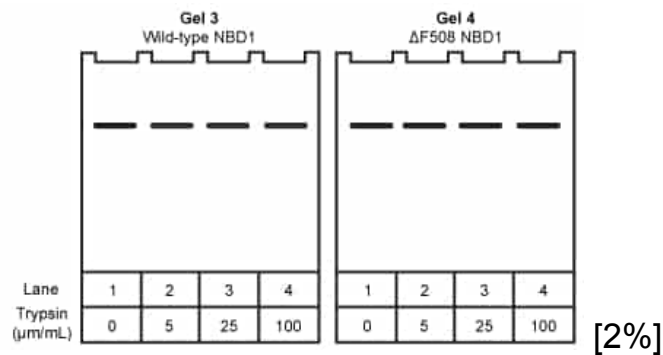
Cystic fibrosis (CF) arises from recessively inherited mutations in the *CFTR* gene. The most common mutation is the deletion of three base-pairs that removes a phenylalanine residue at position 508 in NBD1 ($\Delta F508$). This mutation causes the protein to fold incorrectly, making it more susceptible to degradation by proteases such as trypsin. The resulting decrease in CFTR abundance inhibits chloride ion transport, leading to thickening of normally thin secretions and increased risk of life-threatening pulmonary infections.

If wild-type and $\Delta F508$ NBD1 were isolated and incubated with increasing concentrations of the protease trypsin, which of the following SDS-PAGE gels would be consistent with the increased susceptibility of $\Delta F508$ NBD1 to degradation?

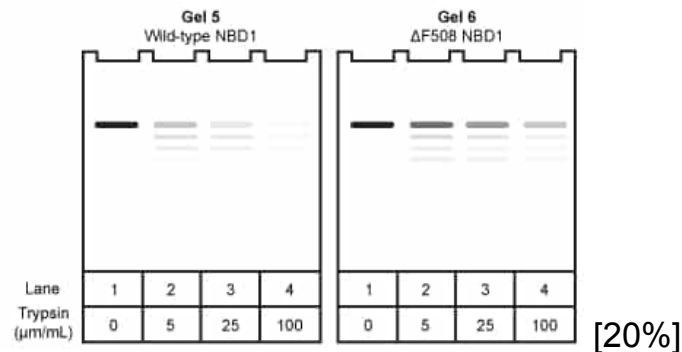
✔ ☒ A.



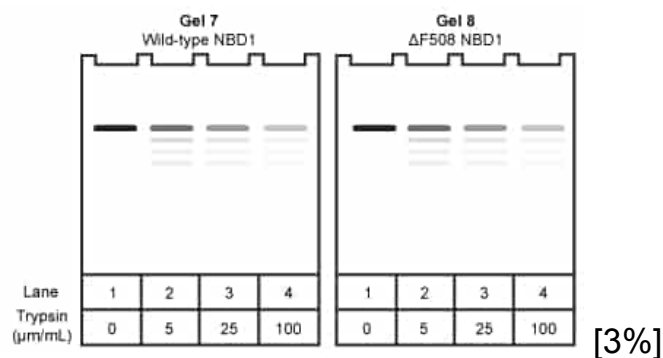
☐ B.



☐ C.



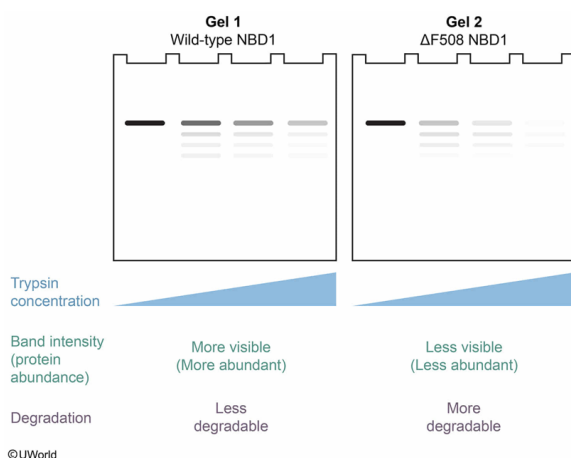
☐ D.



Correct

72% answered correctly

Explanation:



Proteases (eg, trypsin, pepsin) **degrade proteins into smaller fragments** by hydrolyzing the peptide bonds that link amino acids in the polypeptide chain. Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (**SDS-PAGE**) is used to separate proteins by **size (molecular weight)**. Smaller proteins or protein fragments travel through the gel matrix toward the positive anode faster than larger proteins, creating

lanes of size-separated protein bands that are then stained for visualization.

The passage states that the $\Delta F508$ form of CFTR is incorrectly folded, making it more susceptible to degradation by proteases (note that the passage does *not* state that this increased susceptibility depends on the identity of the protease, so any protease may be used). Therefore, the $\Delta F508$ variant undergoes more degradation than the wild-type form when a given concentration of a protease such as trypsin is added. In SDS-PAGE, this appears as a decreased band intensity for each fragment size at a given trypsin concentration.

Therefore, compared to wild-type NBD1, $\Delta F508$ NBD1 should show reduced staining of protein bands in each lane. The correct set of gels is shown in Choice A.

(Choice B) Gels 3 and 4 are similar to each other and show no protein degradation. Trypsin may have been omitted as an experimental error.

(Choice C) Gel 6 ($\Delta F508$ NBD1) has more protease-resistant fragments than Gel 5 (wild-type NBD1). This result would suggest that $\Delta F508$ NBD1 is *less* susceptible to degradation than the wild-type.

(Choice D) Gels 7 and 8 show a similar number of protease-resistant fragments. This indicates similar susceptibility to degradation of both the wild-type and $\Delta F508$ NBD1.

Educational objective:

Proteases degrade proteins into fragments by hydrolyzing the peptide bonds that link amino acids in the polypeptide chain. For SDS-PAGE of a degradable protein, treatment with increasing protease concentrations is correlated with decreased band intensity (abundance) of protease-resistant fragments per lane.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402101
System:
Amino Acids and Proteins